```
!pip install kagglehub
!pip install rouge-score
```

```
Requirement already satisfied: kagglehub in /usr/local/lib/python3.12/dist-packages (0.3.13)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from kagglehub) (25.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from kagglehub) (6.0.3)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kagglehub) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from kagglehub) (4.67.1)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (3
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->kagglehub) (2026.1.
Collecting rouge-score
  Downloading rouge_score-0.1.2.tar.gz (17 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: absl-py in /usr/local/lib/python3.12/dist-packages (from rouge-score) (1.4.0)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (from rouge-score) (3.9.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from rouge-score) (2.0.2)
Requirement already satisfied: six>=1.14.0 in /usr/local/lib/python3.12/dist-packages (from rouge-score) (1.17.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk->rouge-score) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk->rouge-score) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk->rouge-score) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk->rouge-score) (4.67.1)
Building wheels for collected packages: rouge-score
  Building wheel for rouge-score (setup.py) ... done
  Created wheel for rouge-score: filename=rouge_score-0.1.2-py3-none-any.whl size=24934 sha256=da239215216f84d3ab172e96b35aaa4be
  Stored in directory: /root/.cache/pip/wheels/85/9d/af/01feefbe7d55ef5468796f0c68225b6788e85d9d0a281e7a70
Successfully built rouge-score
Installing collected packages: rouge-score
Successfully installed rouge-score-0.1.2
```

```
!pip install transformers datasets rouge-score accelerate
```

```
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers) (2.32.4)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.2
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.1)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025
Requirement already satisfied: absl-py in /usr/local/lib/python3.12/dist-packages (from rouge-score) (1.4.0)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (from rouge-score) (3.9.1)
Requirement already satisfied: six>=1.14.0 in /usr/local/lib/python3.12/dist-packages (from rouge-score) (1.17.0)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.9.0+cu126)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.34
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformer
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers) (2.
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers) (202
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.14.6
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.6
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->a
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->a
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->acce
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->acce
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->acce
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->ac
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->a
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->a
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->ac
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelera
```

```
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspe
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=2.0.0
```

```
import os
import kagglehub
import re
import pandas as pd
import torch
from torch.utils.data import Dataset, DataLoader
from transformers import (
    BartForConditionalGeneration,
    BartTokenizer
)
from rouge_score import rouge_scorer
from torch.optim import AdamW
```

```
path = kagglehub.dataset_download(
    "gowrishankarp/newspaper-text-summarization-cnn-dailymail"
)

print("Dataset path:", path)
files = os.listdir(path)
print("Number of files:", len(files))
```

```
Using Colab cache for faster access to the 'newspaper-text-summarization-cnn-dailymail' dataset.
Dataset path: /kaggle/input/newspaper-text-summarization-cnn-dailymail
Number of files: 1
```

```
data_dir = os.path.join(path, "cnn_dailymail")

train_file = os.path.join(data_dir, "train.csv")
val_file   = os.path.join(data_dir, "validation.csv")
test_file  = os.path.join(data_dir, "test.csv")

print(os.listdir(data_dir))
import pandas as pd

train_df = pd.read_csv(train_file).head(5000)
val_df   = pd.read_csv(val_file).head(1000)
test_df  = pd.read_csv(test_file).head(1000)
```

```
['validation.csv', 'train.csv', 'test.csv']
```

```
import re
import unicodedata
import nltk
from nltk.tokenize import sent_tokenize

!pip install -q nltk

# Download tokenizer data
nltk.download("punkt")
nltk.download("punkt_tab") # Added to resolve LookupError

# -------------------------------
# BART-aware text cleaning
# -------------------------------
def clean_text_bart(text):
    if not isinstance(text, str):
        return ""

    # 1. Unicode normalization
    text = unicodedata.normalize("NFKC", text)

    # 2. Lowercase (optional but safe for CNN-style data)
    text = text.lower()

    # 3. Remove HTML/XML tags
```

```
    text = re.sub(r"<.*?>", " ", text)

    # 4. Remove URLs and emails
    text = re.sub(r"http\S+|www\S+", " ", text)
    text = re.sub(r"\S+@\S+", " ", text)

    # 5. Normalize repeated punctuation
    text = re.sub(r"([.!?]){2,}", r"\1", text)
    text = re.sub(r"[,;:]{2,}", ",", text)

    # 6. Remove non-ASCII artifacts
    text = re.sub(r"[^\x00-\x7F]+", " ", text)

    # 7. Normalize whitespace
    text = re.sub(r"\s+", " ", text).strip()

    # 8. Sentence-level deduplication
    sentences = sent_tokenize(text)
    seen = set()
    cleaned_sentences = []
    for s in sentences:
        if s not in seen and len(s.split()) > 3:
            cleaned_sentences.append(s)
            seen.add(s)

    return " ".join(cleaned_sentences)

# -------------------------------
# Apply cleaning
# -------------------------------
for df in [train_df, val_df, test_df]:
    df["article"] = df["article"].apply(clean_text_bart)
    df["highlights"] = df["highlights"].apply(clean_text_bart)

print("Text cleaning completed.")
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
Text cleaning completed.
```

```
MODEL_NAME = "facebook/bart-large-cnn"

tokenizer = BartTokenizer.from_pretrained(MODEL_NAME)
model = BartForConditionalGeneration.from_pretrained(MODEL_NAME)

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
model.to(DEVICE)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set i
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

vocab.json: 899k/? [00:00<00:00, 19.3MB/s]

merges.txt: 456k/? [00:00<00:00, 24.1MB/s]

tokenizer.json: 1.36M/? [00:00<00:00, 33.3MB/s]

config.json: 1.58k/? [00:00<00:00, 38.5kB/s]

model.safetensors: 100% 1.63G/1.63G [00:18<00:00, 165MB/s]

generation_config.json: 100% 363/363 [00:00<00:00, 12.4kB/s]

```
BartForConditionalGeneration(
  (model): BartModel(
    (shared): BartScaledWordEmbedding(50264, 1024, padding_idx=1)
    (encoder): BartEncoder(
      (embed_tokens): BartScaledWordEmbedding(50264, 1024, padding_idx=1)
      (embed_positions): BartLearnedPositionalEmbedding(1026, 1024)
      (layers): ModuleList(
        (0-11): 12 x BartEncoderLayer(
          (self_attn): BartAttention(
            (k_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (v_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (q_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (out_proj): Linear(in_features=1024, out_features=1024, bias=True)
          )
          (self_attn_layer_norm): LayerNorm((1024,), eps=1e-05, elementwise_affine=True)
          (activation_fn): GELUActivation()
          (fc1): Linear(in_features=1024, out_features=4096, bias=True)
          (fc2): Linear(in_features=4096, out_features=1024, bias=True)
          (final_layer_norm): LayerNorm((1024,), eps=1e-05, elementwise_affine=True)
        )
      )
      (layernorm_embedding): LayerNorm((1024,), eps=1e-05, elementwise_affine=True)
    )
    (decoder): BartDecoder(
      (embed_tokens): BartScaledWordEmbedding(50264, 1024, padding_idx=1)
      (embed_positions): BartLearnedPositionalEmbedding(1026, 1024)
      (layers): ModuleList(
        (0-11): 12 x BartDecoderLayer(
          (self_attn): BartAttention(
            (k_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (v_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (q_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (out_proj): Linear(in_features=1024, out_features=1024, bias=True)
          )
          (activation_fn): GELUActivation()
          (self_attn_layer_norm): LayerNorm((1024,), eps=1e-05, elementwise_affine=True)
          (encoder_attn): BartAttention(
            (k_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (v_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (q_proj): Linear(in_features=1024, out_features=1024, bias=True)
            (out_proj): Linear(in_features=1024, out_features=1024, bias=True)
          )
          (encoder_attn_layer_norm): LayerNorm((1024,), eps=1e-05, elementwise_affine=True)
          (fc1): Linear(in_features=1024, out_features=4096, bias=True)
          (fc2): Linear(in_features=4096, out_features=1024, bias=True)
          (final_layer_norm): LayerNorm((1024,), eps=1e-05, elementwise_affine=True)
        )
      )
      (layernorm_embedding): LayerNorm((1024,), eps=1e-05, elementwise_affine=True)
    )
  )
  (lm_head): Linear(in_features=1024, out_features=50264, bias=False)
)
```

```
class CNNDailyMailDataset(Dataset):
    def __init__(self, articles, summaries, tokenizer, max_input=512, max_output=128):
        self.articles = articles
        self.summaries = summaries
        self.tokenizer = tokenizer
        self.max_input = max_input
        self.max_output = max_output

    def __len__(self):
        return len(self.articles)
```

```
    def __getitem__(self, idx):
        input_enc = tokenizer(
            self.articles[idx],
            truncation=True,
            padding="max_length",
            max_length=self.max_input,
            return_tensors="pt"
        )

        target_enc = tokenizer(
            self.summaries[idx],
            truncation=True,
            padding="max_length",
            max_length=self.max_output,
            return_tensors="pt"
        )

        return {
            "input_ids": input_enc["input_ids"].squeeze(),
            "attention_mask": input_enc["attention_mask"].squeeze(),
            "labels": target_enc["input_ids"].squeeze()
        }
```

```
train_dataset = CNNDailyMailDataset(
    train_df["article"].tolist(),
    train_df["highlights"].tolist(),
    tokenizer
)

val_dataset = CNNDailyMailDataset(
    val_df["article"].tolist(),
    val_df["highlights"].tolist(),
    tokenizer
)

train_loader = DataLoader(train_dataset, batch_size=2, shuffle=True)
val_loader   = DataLoader(val_dataset, batch_size=2)
```

```
optimizer = AdamW(model.parameters(), lr=2e-5)
```

```
def train_epoch(model, loader):
    model.train()
    total_loss = 0

    for batch in loader:
        optimizer.zero_grad()

        input_ids = batch["input_ids"].to(DEVICE)
        attention_mask = batch["attention_mask"].to(DEVICE)
        labels = batch["labels"].to(DEVICE)

        outputs = model(
            input_ids=input_ids,
            attention_mask=attention_mask,
            labels=labels
        )

        loss = outputs.loss
        loss.backward()
        optimizer.step()

        total_loss += loss.item()

    return total_loss / len(loader)

for epoch in range(5):
    loss = train_epoch(model, train_loader)
    print(f"Epoch {epoch+1} | Train Loss: {loss:.4f}")
```

```
Epoch 1 | Train Loss: 0.8841
Epoch 2 | Train Loss: 0.6031
Epoch 3 | Train Loss: 0.4198
```

```
Epoch 4 | Train Loss: 0.2829
Epoch 5 | Train Loss: 0.1934
```

```
scorer = rouge_scorer.RougeScorer(["rouge1", "rouge2", "rougeL"], use_stemmer=True)

def evaluate(model, df, n_samples=100):
    model.eval()
    scores = {"rouge1": [], "rouge2": [], "rougeL": []}

    for i in range(n_samples):
        article = df.iloc[i]["article"]
        reference = df.iloc[i]["highlights"]

        inputs = tokenizer(
            article,
            return_tensors="pt",
            truncation=True,
            max_length=512
        ).to(DEVICE)

        summary_ids = model.generate(
            **inputs,
            max_length=128,
            min_length=40,
            length_penalty=2.0,
            num_beams=4
        )

        generated = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
        rouge = scorer.score(reference, generated)

        for k in scores:
            scores[k].append(rouge[k].fmeasure)

    return {k: sum(v)/len(v) for k, v in scores.items()}

rouge_scores = evaluate(model, test_df)
print(rouge_scores)
```

```
{'rouge1': 0.4116847493642784, 'rouge2': 0.17414742601357924, 'rougeL': 0.2723406276932353}
```

```
def show_examples(model, df, n_examples=3):
    model.eval()

    for i in range(n_examples):
        article = df.iloc[i]["article"]
        reference = df.iloc[i]["highlights"]

        inputs = tokenizer(
            article,
            return_tensors="pt",
            truncation=True,
            max_length=512
        ).to(DEVICE)

        summary_ids = model.generate(
            **inputs,
            max_length=128,
            min_length=40,
            length_penalty=2.0,
            num_beams=4
        )

        generated = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

        print("=" * 80)
        print(f"Example {i + 1}")
        print("-" * 80)
        print("ARTICLE:\n", article[:1000], "...\n")
        print("REFERENCE SUMMARY:\n", reference, "\n")
        print("GENERATED SUMMARY:\n", generated, "\n")
show_examples(model, test_df, n_examples=5)
```

```
================================================================================
Example 1
--------------------------------------------------------------------------------
ARTICLE:
```

```
 ever noticed how plane seats appear to be getting smaller and smaller? with increasing numbers of people taking to the skies,

REFERENCE SUMMARY:
 experts question if packed out planes are putting passengers at risk . u.s consumer advisory group says minimum space must be

GENERATED SUMMARY:
 experts say that shrinking space on planes is putting our health and safety in danger . a consumer advisory group said that wh

================================================================================
Example 2
--------------------------------------------------------------------------------
ARTICLE:
 a drunk teenage boy had to be rescued by security after jumping into a lions' enclosure at a zoo in western india. rahul kuma

REFERENCE SUMMARY:
 drunk teenage boy climbed into lion enclosure at zoo in west india . rahul kumar, 17, ran towards animals shouting 'today i k

GENERATED SUMMARY:
 rahul kumar, 17, scaled fence at kamla nehru zoological park in ahmedabad . intoxicated teenager ran towards the lions, shout

================================================================================
Example 3
--------------------------------------------------------------------------------
ARTICLE:
 dougie freedman is on the verge of agreeing a new two-year deal to remain at nottingham forest. freedman has stabilised forest

REFERENCE SUMMARY:
 nottingham forest are close to extending dougie freedman's contract . the forest boss took over from former manager stuart pea

GENERATED SUMMARY:
 dougie freedman is set to sign a new two-year deal at nottingham forest . freedman has stabilised the club since taking over f

================================================================================
Example 4
--------------------------------------------------------------------------------
ARTICLE:
 liverpool target neto is also wanted by psg and clubs in spain as brendan rodgers faces stiff competition to land the fiorent

REFERENCE SUMMARY:
 fiorentina goalkeeper neto has been linked with liverpool and arsenal . neto joined firoentina from brazilian outfit atletico

GENERATED SUMMARY:
 firoentina goalkeeper neto is wanted by a number of top european clubs, according to the brazilian's agent stefano castagna .

================================================================================
Example 5
--------------------------------------------------------------------------------
```